

Test Cases

Test Case ID	Input Scenario	Expected Module	Expected Result	Status
TC001	HR login with valid credentials	Authentication	HR Dashboard opens successfully	Pass
TC002	Employee login with valid credentials	Authentication	Employee Dashboard opens successfully	Pass
TC003	Create new employee onboarding request	Onboarding Agent	Employee record created and workflow starts	Pass
TC004	Upload CSV containing Email, Phone and PAN	PII Detection	Sensitive data detected successfully	Pass
TC005	Generate masked file from uploaded CSV	PII Protection	Masked copy generated successfully	Pass
TC006	Assign Morning Shift to employee	Shift Management	Morning Shift displayed on dashboard	Pass
TC007	Assign Night Shift to employee	Shift Management	Night Shift displayed on dashboard	Pass
TC008	HR sends announcement	Notification Center	Employee receives notification badge	Pass
TC009	Employee uploads Resume PDF	Document Center	Document uploaded successfully	Pass
TC010	Employee asks onboarding question	Employee AI Assistant	Relevant AI response generated	Pass
TC011	HR asks for onboarding summary	HR AI Assistant	Onboarding status report generated	Pass
TC012	Employee views attendance	Attendance Module	Attendance displayed correctly	Pass
TC013	Employee views learning recommendations	Learning Engine	Recommended courses displayed	Pass
TC014	Employee submits feedback	Feedback Module	Feedback stored successfully	Pass
TC015	HR reviews PII risk report	Risk Dashboard	Risk report displayed correctly	Pass

Functional Testing

Authentication

- Verify HR login works correctly.
- Verify Employee login works correctly.
- Verify invalid credentials are rejected.
- Verify logout functionality works.

Employee Onboarding

- Verify new employee creation.
- Verify onboarding workflow execution.
- Verify onboarding checklist generation.
- Verify onboarding completion tracking.

PII Detection & Masking

- Verify CSV files are processed correctly.
- Verify JSON files are processed correctly.
- Verify Email detection.
- Verify Phone Number detection.
- Verify PAN detection.
- Verify Aadhaar detection.
- Verify masked file generation.
- Verify privacy report generation.

Employee Dashboard

- Verify dashboard loads successfully.
- Verify attendance module works.
- Verify shift details display correctly.
- Verify learning recommendations display.
- Verify roadmap section works.
- Verify achievement badges display.
- Verify notifications appear correctly.

HR Dashboard

- Verify employee analytics loads.
- Verify Smart Alert Center works.
- Verify PII Risk Dashboard works.
- Verify Pending Approvals section works.
- Verify AI Task Prioritization works.

Chatbot Testing

- Verify Employee AI Assistant responds correctly.
- Verify HR AI Assistant responds correctly.

- Verify fallback responses work.
- Verify onboarding FAQs work.

Database Storage Testing

- Verify SQLite database is created.
- Verify employee records are stored.
- Verify attendance records are stored.
- Verify onboarding records are stored.
- Verify notification records are stored.
- Verify PII reports are stored.

Agent Workflow Testing

Onboarding Agent

Verify:

- Create Account step executes.
- Request Laptop step executes.
- Assign License step executes.
- Welcome Notification step executes.
- Workflow completion status updates.

Dashboard Testing

- Verify Employee Dashboard loads successfully.
- Verify HR Dashboard loads successfully.
- Verify charts load correctly.
- Verify sidebar navigation works.
- Verify profile panel opens correctly.
- Verify notification panel opens correctly.

Performance Testing

- Process 50 employee records within 5 seconds.
- Process 100 employee records without application crash.
- Process 50 PII records successfully.
- Notification delivery success rate: 100%.
- Database insertion success rate: 100%.

Error Handling Testing

- Missing employee name.
- Missing employee email.
- Invalid login credentials.
- Empty CSV file upload.
- Invalid JSON format.
- Unsupported document upload.

- Missing chatbot API key.
- Database connection failure.
- Invalid report ID access.
- Unauthorized dashboard access.

Security Testing

- Employee cannot access HR Dashboard.
- Protected routes require login.
- Uploaded PII is masked before download.
- Session expires after logout.
- Unauthorized file access is blocked.

Expected Outcome

- All test cases pass successfully.
- Employee onboarding workflow executes correctly.
- PII data is detected and masked.
- HR and Employee dashboards function properly.
- AI Assistants generate valid responses.
- Notifications, attendance, documents, and reports work correctly.